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PI Name:	Bronner-Fraser, Marianne Ph.D.		
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PI Email:	mbronner@gg.caltech.edu	**Fax:**	FY 626-395-7717
PI Organization Type:	UNIVERSITY	**Phone:**	626-395-3355
Organization Name:	California Institute of Technology		
PI Address 1:	Division of Biology, Mail Stop 139-74		
PI Address 2:	1200 East California Boulevard		
PI Web Page:			
City:	Pasadena	**State:**	CA
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Task Description:	The neural crest and the cranial ectodermal placodes give rise to many of the defining characters of vertebrates, including a well-defined head with teeth and paired sensory organs. Both are migratory, form at the border between neural plate and epidermis, and generate some identical derivatives such as sensory neurons. To explore the events that led to the evolution of vertebrate features, this research will examine: (1) the evolutionary origin of the neural crest and placodes and (2) the interrelationship and evolutionary divergence of these two important embryonic structures. We aim to look for evolutionary precursors to neural crest and placodes in the invertebrate chordate, amphioxus, and to study neural crest and placodes in the primitive vertebrate, lamprey. We will utilize vital dye labeling to examine whether amphioxus has migratory cells that resemble neural crest/placodal cells in vertebrates. We will correlate the sources of migratory cells with gene expression domains of homologs to vertebrate molecular markers for neural crest cells and		

	placodes. In addition, we will map neural crest migratory pathways and derivatives in the lamprey. These represent questions of major importance in the evolution of the vertebrate body plan. Task Objective: Examine evolutionary origin of vertebrate neural crest and placodes, with emphasis on ear. Task Significance: Important for understanding vertebrate origins as well as evolution of vestibular system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the origin of life on earth is critical to understanding if life may have evolved elsewhere in the universe. This research promises to be highly informative regarding the origin of vertebrate body plan. Neural crest cells and placodes are unique to vertebrates and are some of their most defining characteristics. However, little is known about how these tissue types evolved. By understanding their evolutionary origin from non-vertebrate chordates, we hope to better discern how the vertebrate body plan arose.
Task Progress:	1. Ancient role for vertebrate SoxE genes in pharyngeal arch development. Neural crest cells are a defining characteristic that sets vertebrates apart from non- vertebrate chordates1 and were of prime importance in the evolutionary origin of the vertebrate body plan2. To understand neural crest evolution, we explored molecular mechanisms underlying craniofacial development in the basal jawless vertebrate, lamprey. Here, we have isolated a lamprey SoxE related gene that we name PmSoxE1. This gene is expressed in migrating cranial neural crest cells and their derivatives in the pharyngeal arches. In jawed vertebrates, SoxE members (Sox8, Sox9, and Sox10) are important transcriptional regulators of neural crest development3 and loss of Sox9 causes abnormal cartilage formation. Using in situ hybridization, cell labeling, and morpholino-induced protein knock-down, we demonstrate that PmSoxE1 is expressed in prechondrocytes of the branchial arches and is required for development of the neural crest and pharyngeal arches. Our results demonstrate that function of this transcriptional regulator can be traced to the lamprey-gnathostome common ancestor, and that a SoxE gene was important in the evolution of vertebrate chondrogenesis in spite of morphological, positional, and structural differences seen in the cartilages of larval lampreys and gnathostomes. 2. Amphioxus SoxE suggests subfunctional of SoxE genes occurred in the vertebrate lineage. A comparison of gene numbers among chordates indicate that 1-2 whole genome duplications occured near the time of vertebrate origins. It has been suggested that these duplications potentiated the evolution of vertebrates by supplementing the genetic toolkit of the vertebrate ancestor. To test this idea, we are studying the function evolution of SoxE genes in the chordate lineage. The vertebrate SoxE genes, Sox8,9,and 10 perform critical and largely non-redundant roles the development of the neural crest-- an embryonic tissue unique to vertebrate embryos. Basal chordates, such as amphioxus and ascidians possess only single SoxE genes which do not appear to be expressed cells with neural crest-like properties. Using overexpression and gene rescue strategies in frog and zebrafish, we are assaying the ability of basal chordate SoxE genes to function like vertebrate SoxE paralogs in neural crest cells. Preliminary data suggest that while amphioxus SoxE is capable of up-regulating pan-neural crest marker genes, it does not drive neural crest cells towards a melanocyte fate, as vertebrate Sox10 genes do. This provides some evidence for a neo-functionalization associated with the neural crest evolution. 3. Spalt4 triggers ectodermal invagination to form sensory placodes Vertebrate placodes are regions of thickened head ectoderm that contribute to paired sense organs and sensory ganglia. Here, we show that the transcription factor spalt4 is expressed broadly in chick preplacodal epiblast and then resolves to otic and olfactory placodes and their derivatives. Misexpression of spalt4 in non-placodal ectoderm by electroporation results in formation of ectopic vesicles that express the otic markers Pax2, Sox 10 and Tbx1 in the vicinity of the ear, and the lens/olfactory marker Pax6 in more rostral ectoderm. Furthermore, trunk ectoderm that normally does not form placodes, ingresses after misexpression of spalt4. Excess Spalt4 in the otic placode causes severe malformations and alterations in region-specific markers. Conversely, loss of spalt4 function using a dominant negative construct results in significant reductions in the otic vesicle. These results suggest that spalt4 is both necessary and sufficient for ear development, initiating invagination/ingression of ectoderm and region-specific gene regulatory networks. 4. Subcontract—Dr. Linda Holland This grant has supported one graduate student, Jr-Kai Yu, who obtained the PhD degree in March, 2005. To date, he is an author on nine papers, five of them as first author. The goal of our NASA-supported research has been to understand changes at the gene level that resulted in the evolution of vertebrate neural crest, which contributes essential elements to such important vertebrate features as the inner ear. Invertebrate chordates (tunicates and amphioxus) lack neural crest even though, like vertebrates, they have a dorsal, hollow nerve cord, which expresses many of the same genes in similar patterns as in vertebrates. To understand the innovations at the gene level that endowed cells at the edges of the neural plate with the ability to migrate away from the neural tube and to differentiate into such structures as the cartilage of the inner ear, we have focused on the FoxD genes in the forkhead family of transcription factors. During the transition from invertebrate to vertebrate chordates, large-scale gene duplications occurred resulting in 5 FoxD genes in vertebrates compared to only one in the invertebrate chordate amphioxus. Of these, only FoxD3 is expressed in premigratory neural crest cells at the edges of the neural plate as well as in migratory neural crest and has been shown to have a role initiation of neural crest cell migration and in differentiation of neural crest cells. Our work supported under this NASA grant showed that AmphiFoxD is not expressed in the edges of the neural plate in amphioxus (Yu et al., 2002). Therefore, we asked how FoxD genes have changed subsequent to gene duplication in order to allow new roles in neural crest. We have considered two possibilities. One is that new cis-regulatory elements directing expression to the edges of the neural plate evolved in the vertebrate FoxD3 gene subsequent to duplication of the FoxD genes in the vertebrates. The second is that the FoxD3 protein changed as well allowing the FoxD3 gene to perform new functions in cell migration and differentiation. For the first objective, we cloned 6.3 kb of upstream AmphiFoxD regulatory sequence into a lacZ reporter vector and expressed it in amphioxus. Methods for microinjection of amphioxus eggs that we developed were published in “Methods in Cell Biology (Holland and Yu, 2004). This work, (Yu et al., 2004) was published in Developmental Biology. In brief, we showed that this 6.3 kb region of upstream regulatory DNA is sufficient for directing expression to the three tissues that normally express the gene—the notochord, somites and anterior nerve cord. In addition, we identified a 4.7 kb region that was sufficient for directing expression to the notochord. In addition, we suggested in this paper that a cluster of binding sites for the notochord marker, brachyury, might be essential for notochordal expression.

To further test if the putative Brachyury binding sites located in the 4.7 kb region are necessary for tissue expression, especially for the notochord expression, we generated the delta Bra construct which has a deletion eliminating the putative Brachyury/Tbx5 binding sites from the original -6.3 kb construct. As we expected, the delta Bra construct directed expression to somites and nerve chord but not to the notochord. This result indicates that the putative Brachyury/Tbx5 binding sites are essential for reporter gene constructs to be expressed in the notochord, and, thus, that this DNA region is a notochord enhancer.

In addition, we had suggested that the region with 2.5 repeats of a 268 bp motif might be an autoregulatory enhancer. Recently, we have found that it is indeed an autoregulatory enhancer. We also generated the delta Repeat construct to test if the 2.5 tandem repeat region is important for tissue specific expression. This delta Repeat construct has a deletion eliminating the entire repeat region from the -6.3 kb construct. The result showed that this delta Repeat construct only rarely induces some weak expression in the somite and notochord but never any expression in the nerve chord. This suggests that this tandem repeat region is probably a general enhancer region that is essential for inducing/maintaining AmphiFoxD expression in all major domains. In collaboration with Dr. Mark Reedy (Creighton University), we have cloned the 6.3 kb construct into an expression vector suitable for the chick and electroporated it into the neural tube of chick embryos. As we did not see any expression with this construct, we conclude that cis-regulatory elements capable of directing expression to neural crest probably evolved in the FoxD3 gene subsequent to duplication of the vertebrate FoxD genes, although, of course, we cannot rule out the possibility that a region of the FoxD3 gene either upstream or downstream of the one we tested possesses such regulatory elements. These experiments are being continued in collaboration with Marianne Bronner-Fraser.

To test the possibility that the function of the FoxD protein has also changed to allow a new role in promoting neural crest, in collaboration with Marianne Bronner-Fraser, we overexpressed AmphiFoxD in the neural plate of eggs of the frog, Xenopus laevis and compared it to vertebrate FoxD1 and FoxD3. Injection of a construct containing the DNA binding domain of AmphiFoxD but with the transactivation domain substituted with Gal4, has no effect on the neural plate. However, injection of a construct with the DNA binding domain of AmphiFoxD linked to the engrailed repressor domain expands the neural plate as does injection of AmphiFoxD, Xenopus FoxD1 and Xenopus FoxD3. Similarly, overexpression of amphioxus FoxD in the Xenopus neural plate promotes expansion of the neural plate. We conclude that amphioxus FoxD protein acts as a repressor and that it has similar function as both vertebrate FoxD3 and FoxD1.

Recently, we have been taking advantage of the amphioxus genome project to identify additional genes of potential importance to the evolution of neural crest and placodes. These include neural cadherin and members of the fibroblast growth factor family.

In summary, our NASA-supported work has resulted in five papers in 2003/2004. These include a phylogenetic analysis of FoxD genes (Mazet et al., 2003), a methods paper describing techniques for expression of reporter constructs in amphioxus eggs (Holland and Yu, 2004) and a paper describing the tissue-specific expression of reporter constructs in amphioxus eggs (Yu et al., 2004). This work has shown that evolution of neural crest probably involved the evolution of new cis-regulatory elements in the FoxD3 gene after gene duplication. These results have shed light on the mechanism of evolution of neural crest and on the evolution of new structures generally.

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